

## STRAVA FITNESS

### SQL REPORT (Data Loading & Cleaning)

This document details the complete SQL-based process used to import, clean, and validate the Strava Fitness dataset. The raw data consisted of 18 separate CSV files, which were loaded into a MySQL database named strava.

- **Data Loading** : A total of 18 tables were created to house the data. The LOAD DATA LOCAL INFILE command was used for efficient bulk insertion from the CSV files. A critical part of this process was using session variables (e.g., @csv\_ActivityDate) and the STR\_TO\_DATE() function to correctly parse and format date, time, and boolean values from the raw text. Two example of such task is :

```
create table dailyactivity(Id bigint, ActivityDate datetime, TotalSteps int, TotalDistance float, TrackerDistance float, LoggedAct
LOAD DATA LOCAL INFILE 'C:/Users/Sumit Sinha/Desktop/Strava Fitness/dailyActivity_merged.csv' INTO TABLE dailyactivity
FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 ROWS (Id, @csv_ActivityDate, TotalSteps, TotalDistance, TrackerDistance
SET ActivityDate=STR_TO_DATE(@csv_ActivityDate, '%m/%d/%Y');
SELECT * FROM dailyactivity;

create table hourlycalories(Id bigint, ActivityHour datetime, Calories int);
LOAD DATA LOCAL INFILE 'C:/Users/Sumit Sinha/Desktop/Strava Fitness/hourlycalories_merged.csv' INTO TABLE hourlycalories
FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 ROWS (Id, @csv_ActivityHour, Calories)
SET ActivityHour = STR_TO_DATE(@csv_ActivityHour, '%m/%d/%Y %l:%i:%s %p');
SELECT * FROM hourlycalories;
```

The same methodology was used to create and load all 18 tables:

- dailyactivity
  - dailycalories
  - dailyintensity
  - dailysteps
  - heartrateseconds
  - hourlycalories
  - hourlyintensities
  - hourlysteps
  - minutecaloriesnarrow
  - minutecalorieswide
  - minuteintensitiesnarrow
  - minuteintensitieswide
  - minutemetnarrow
  - minutesleepmerged
  - minutestepsnarrow
  - minutestepswide
  - sleepday
  - weightlog
- **Data Cleaning & Validation** : The first step was to validate the number of unique participants (Id) in each table to understand the scope and limitations of the data. This revealed different sample sizes for different features.

- **Duplicate Record Check** : A systematic check for duplicate rows (i.e., multiple records for the same user at the same time) was performed using a GROUP BY query on the primary keys. This check confirmed that 16 of the 18 tables were clean. Duplicates were found in two tables: minutesleepmerged and sleepday. Example of such check is :

```
SELECT Id, ActivityDate, COUNT(*) AS NumOfRecords FROM dailyactivity GROUP BY Id, ActivityDate HAVING COUNT(*) > 1; #0
SELECT Id, ActivityDay, COUNT(*) AS NumOfRecords FROM dailycalories GROUP BY Id, ActivityDay HAVING COUNT(*) > 1; #0
SELECT Id, ActivityDay, COUNT(*) AS NumOfRecords FROM dailyintensity GROUP BY Id, ActivityDay HAVING COUNT(*) > 1; #0
```

- **Duplicate Removal** :

```
CREATE TABLE minutesleepmerged_clean LIKE minutesleepmerged;
INSERT INTO minutesleepmerged_clean SELECT DISTINCT * FROM minutesleepmerged;
SELECT Id, date, COUNT(*) AS NumOfRecords FROM minutesleepmerged_clean GROUP BY Id, date HAVING COUNT(*) > 1;
CREATE TABLE sleepday_clean LIKE sleepday;
INSERT INTO sleepday_clean SELECT DISTINCT * FROM sleepday;
SELECT Id, SleepDay, COUNT(*) AS NumOfRecords FROM sleepday_clean GROUP BY Id, SleepDay HAVING COUNT(*) > 1;
select count(*) from minutesleepmerged_clean;
```

- **Checking for Null Values** : The data was checked for NULL values. This process helped identify "non-wear" days. The result came out to be 77 days.

```
SELECT * FROM weightlog WHERE Fat IS NULL OR BMI IS NULL;
SELECT COUNT(*) FROM dailyactivity WHERE TotalSteps = 0; #77
```

- **Illogical Data Checks** : The data was finally checked for any illogical or impossible data. This process helped us in identifying whether there is any data entry errors.

```
#Checking for any illogical data :
SELECT * FROM sleepday_clean WHERE TotalMinutesAsleep>TotalTimeInBed;
SELECT Id, ActivityDate, (VeryActiveMinutes+FairlyActiveMinutes+LightlyActiveMinutes+SedentaryMinutes) AS TotalMinutes
FROM dailyactivity WHERE (VeryActiveMinutes+FairlyActiveMinutes+LightlyActiveMinutes+SedentaryMinutes)>1440;
SELECT * FROM weightlog WHERE WeightKg<=0 OR WeightKg>200;

#Checking for negative values :
SELECT min(TotalDistance), max(TotalDistance), min(TotalSteps), max(TotalSteps), min(LoggedActivitiesDistance), min(SedentaryActiveDistance),
max(SedentaryActiveDistance), min(SedentaryMinutes), max(SedentaryMinutes), min(Calories), max(Calories) FROM dailyactivity;

SELECT id, SedentaryMinutes, count(*) AS 'Sedentary Days (=> 1440 mins)' FROM dailyactivity WHERE SedentaryMinutes >= 1440 GROUP BY id, S
```

- **Conclusion** : Following these loading and cleaning steps, the strava database is now validated, de-duplicated, and internally consistent. The dataset is confirmed to have 33 core participants, with smaller cohorts for sleep (24), heart rate (14), and weight (8). The database is now considered clean and is ready for the analysis and visualization phase in Python and Power BI.

-Sumit Sinha